# What Moves Oil? Financial and Economic Conditions Behind Monthly WTI Returns, 2018–2025

FINARTS C01 | Group Case 1: Comprehensive Financial Regression Analysis

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## Executive Summary

**Problem.** Philippine firms that buy fuel, such as distributors, airlines, power producers and transport operators, are exposed to swings in world crude oil prices that they cannot control. To manage that exposure they need to know which financial and economic conditions move with oil. This report asks: *over January 2018 – December 2025, which global financial conditions and which Philippine macro-financial conditions are associated with monthly WTI crude oil log returns, and how strongly?*

**Approach.** We regress 96 monthly WTI log returns on the eight candidate regressors in the case brief: S&P 500 returns, the US dollar index, VIX, gold, US 10-year yield changes, Philippine CPI inflation, USD/PHP and Philippine manufacturing growth. We then test the results, compare specifications, run diagnostics, correct the problems found, and check the final model with robust MM-regression.

**Key findings.**

- Oil returns are tied to **global** conditions. In the final model, a 10 bp rise in the US 10-year yield goes with a 1.35 pp higher monthly WTI return, a 1% S&P 500 gain with 0.61 pp, and a 1% stronger dollar with 1.35 pp *lower* (HAC p-values <0.001, 0.001 and 0.046).
- **Philippine conditions add no explanatory power** (joint F-test p = 0.169). The peso's apparent link to oil reflects oil weakening the peso, not the reverse.
- The initial model was distorted by **March and May 2020** (-78% and +62%). Event dummies with Newey-West standard errors fix every diagnostic failure. A robust MM-estimator independently gives those same two months zero weight and reproduces the final coefficients.

**Recommendation.** Philippine oil users should base oil-risk monitoring and hedging on **global indicators** (US yields, equities and the dollar) rather than domestic data. Because oil increases tend to coincide with a weaker peso, they should **hedge oil and USD/PHP together**. They should also protect against **tail events** with options and stress tests, since the largest moves were not explained by any financial variable.

## 1 Introduction

**Decision context.** The Philippines imports nearly all of its crude oil and refined fuel. For a domestic fuel buyer, a rise in world oil prices raises costs directly, and through the exchange rate it also weakens

the peso in which those costs are paid. Treasury and risk committees must decide what to monitor and how to hedge. That decision depends on whether oil price movements are linked to observable financial conditions, and whether those conditions are global or domestic.

**Empirical research question.** The general question in the brief, "which financial and economic conditions are associated with WTI monthly log returns?", is refined into two testable questions:

1. Which **global** financial conditions (equity returns, the dollar, volatility, gold and US long-term yields) are significantly associated with contemporaneous monthly WTI log returns, and how large are those associations economically?
2. Once global conditions are controlled for, do **Philippine** conditions (inflation, the peso and manufacturing growth) add any explanatory power? A small, price-taking oil importer should find that they do not.

**Structure.** Section 2 describes the data. Sections 3–5 estimate, test and compare the initial specifications. Section 6 presents the diagnostics, Section 7 the remedies, and Section 8 the robust regression. Sections 9–10 set out the financial interpretation and the recommendation. Each section is produced by its own R Markdown file (`0_` to `8_`), and all tables and figures are generated in R.

# 2 Data and Variables

## 2.1 Dataset and unit of observation

The unit of observation is one **calendar month**. The sample runs from January 2018 to December 2025, giving $T = 96$ monthly observations. Data for 2016–2017 is downloaded only so that the January 2018 returns and year-on-year rates can be computed. Every series is downloaded by the R workflow (`0_data_collection_and_library.Rmd`) and cached in `data/raw_monthly.csv`. The final analysis dataset is saved as `data/analysis_dataset.csv`.

Table 1: Variable definitions, transformations and sources

| Variable | Code | Construction | Source | Expected sign |
|---|---|---|---|---|
| WTI crude oil log return (dependent) | WTI | 100 x log change of month-end spot | FRED DCOILWTICO (EIA) | |
| S&P 500 return | SP500 | 100 x log change of month-end level | FRED SP500 | + |
| USD index return | USD | 100 x log change of month-end level | FRED DTWEXBGS | - |
| VIX | VIX | Monthly average of daily close | FRED VIXCLS | - |
| Gold return | Gold | 100 x log change of month-end price | COMEX gold futures (Yahoo GC=F) | +/- |
| US 10Y yield change | d10Y | First difference of month-end yield (pp) | FRED DGS10 | + |
| PH CPI inflation | PH_CPI | YoY % change, headline CPI (2018=100) | PSA OpenSTAT | +/0 |
| USD/PHP change | USDPHP | 100 x log change of month-end rate | Yahoo PHP=X | +/0 |
| PH manufacturing growth | PH_Mfg | YoY % change, MISSI VoPI (2018=100) | PSA OpenSTAT (MISSI) | +/0 |

**Dependent variable.** The dependent variable is the monthly log return on West Texas Intermediate crude oil, $r_t = 100\,[\ln P_t - \ln P_{t-1}]$. $P_t$ is the last available daily EIA spot price in month $t$. Using month-end prices follows the case brief, and it also avoids the single negative daily print of 20 April 2020 (–$37.63/bbl), at which a logarithm is undefined. The April 2020 month-end price was positive.

**Explanatory variables.** The regressors follow the case brief and fall into two blocks:

- *Global financial conditions:* equity returns, the trade-weighted dollar, implied volatility, gold and

US long-term yields. Oil is priced in dollars and traded globally, so these are the conditions the literature associates with oil prices (Kilian, 2009; Akram, 2009).

- *Philippine conditions:* inflation, the peso and manufacturing activity. The Philippines is a small, open, net oil-importing economy and a price-taker in world oil markets. These variables are therefore expected to *respond to* oil rather than drive it. Testing that expectation formally is one of the aims of the case.

## 2.2 Data-quality checks

Table 2: Data types and missing observations, January 2018 – December 2025

| Variable | Type | Missing | Non-finite |
|---|---|---|---|
| WTI | numeric | 0 | 0 |
| S&P 500 return | numeric | 0 | 0 |
| USD index return | numeric | 0 | 0 |
| VIX (monthly avg.) | numeric | 0 | 0 |
| Gold return | numeric | 0 | 0 |
| Change in US 10Y yield | numeric | 0 | 0 |
| PH CPI inflation (YoY) | numeric | 0 | 0 |
| USD/PHP change | numeric | 0 | 0 |
| PH manufacturing growth (YoY) | numeric | 0 | 0 |

All series are numeric, and there are no missing or non-finite values in the estimation sample. Weekend and holiday gaps in the daily series are skipped when month-end values are taken, so they cause no missing data. Two data-quality issues still need attention:

1. **Source substitutions.** The brief points to the World Gold Council for gold and to the BSP for USD/PHP. Neither offers a scriptable download, so COMEX front-month gold futures and Yahoo's USD/PHP close are used instead. Both track the official series closely at month-end, and the substitution keeps the workflow fully reproducible.
2. **Base effects in Philippine manufacturing.** The PSA Volume of Production Index fell to about 14–17% of its 2018 level during the 2020 lockdowns. Year-on-year growth therefore ranges from -82.2% to 534.2%, because 2021 is measured against a collapsed 2020 base. The published series also shows a large drop in July 2019. These values are PSA's official figures and are kept as the brief specifies. The diagnostics check whether they distort the regression.

## 2.3 Descriptive statistics

Table 3: Descriptive statistics (monthly, % or percentage points)

| Variable | Mean | SD | Min | Median | Max | Skewness | Kurtosis | AR(1) |
|---|---|---|---|---|---|---|---|---|
| WTI | -0.06 | 13.56 | -78.20 | 0.56 | 61.50 | -1.09 | 16.51 | 0.13 |
| S&P 500 return | 0.98 | 4.77 | -13.37 | 1.94 | 11.94 | -0.59 | 3.22 | -0.14 |
| USD index return | 0.09 | 1.60 | -3.30 | -0.08 | 4.06 | 0.03 | 2.45 | 0.04 |
| VIX (monthly avg.) | 19.73 | 6.76 | 11.06 | 18.34 | 57.74 | 2.33 | 12.24 | 0.66 |
| Gold return | 1.25 | 3.79 | -7.13 | 1.01 | 9.67 | 0.13 | 2.32 | 0.09 |
| Change in US 10Y yield | 0.02 | 0.26 | -0.52 | 0.01 | 0.68 | 0.05 | 2.74 | 0.08 |
| PH CPI inflation (YoY) | 3.83 | 1.91 | 0.49 | 3.72 | 8.68 | 0.59 | 2.71 | 0.96 |
| USD/PHP change | 0.18 | 1.68 | -3.81 | -0.04 | 5.08 | 0.58 | 3.25 | -0.03 |
| PH manufacturing growth (YoY) | 20.85 | 103.29 | -82.17 | 1.55 | 534.25 | 3.73 | 17.21 | 0.76 |

Average WTI returns are close to zero (-0.06% a month), but the series is very volatile (SD 13.56%). It is also strongly non-normal, with a kurtosis of 16.5 against 3 for a normal distribution. This is driven by March 2020 (-78.2%) and May 2020 (61.5%). The return series show little first-order persistence. VIX (AR(1) = 0.66) and PH inflation (0.96) are persistent levels, as expected for a volatility index and a year-on-year rate. PH manufacturing growth has the extreme range described above.

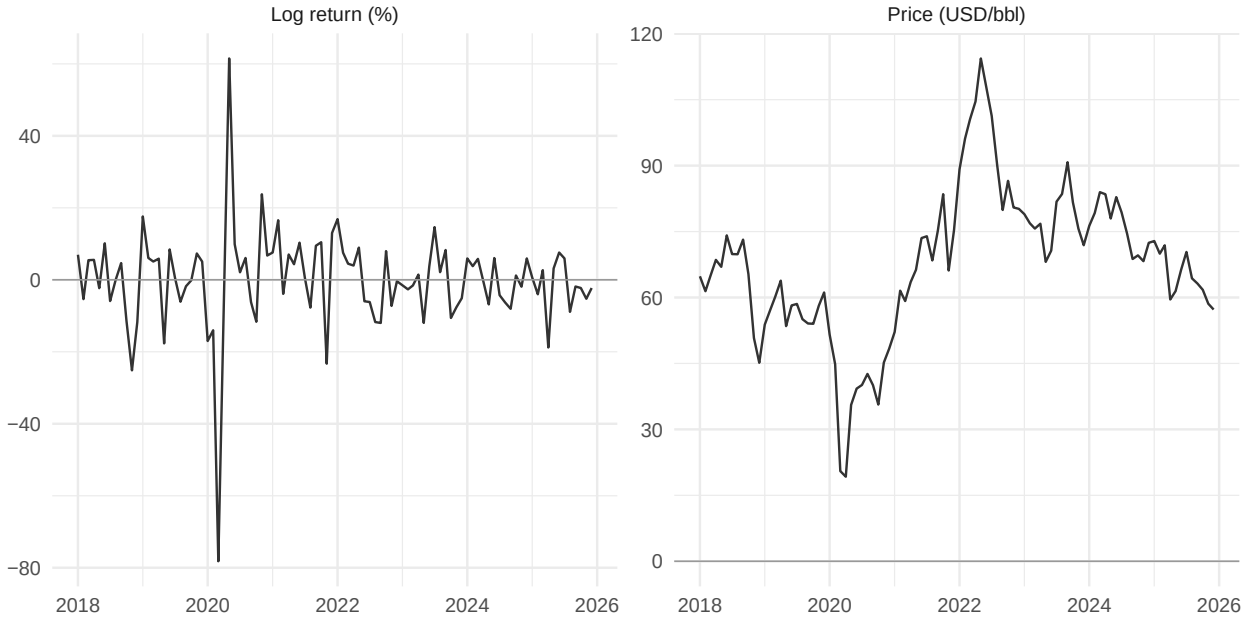



Figure 1: WTI month-end price and monthly log return, 2018–2025

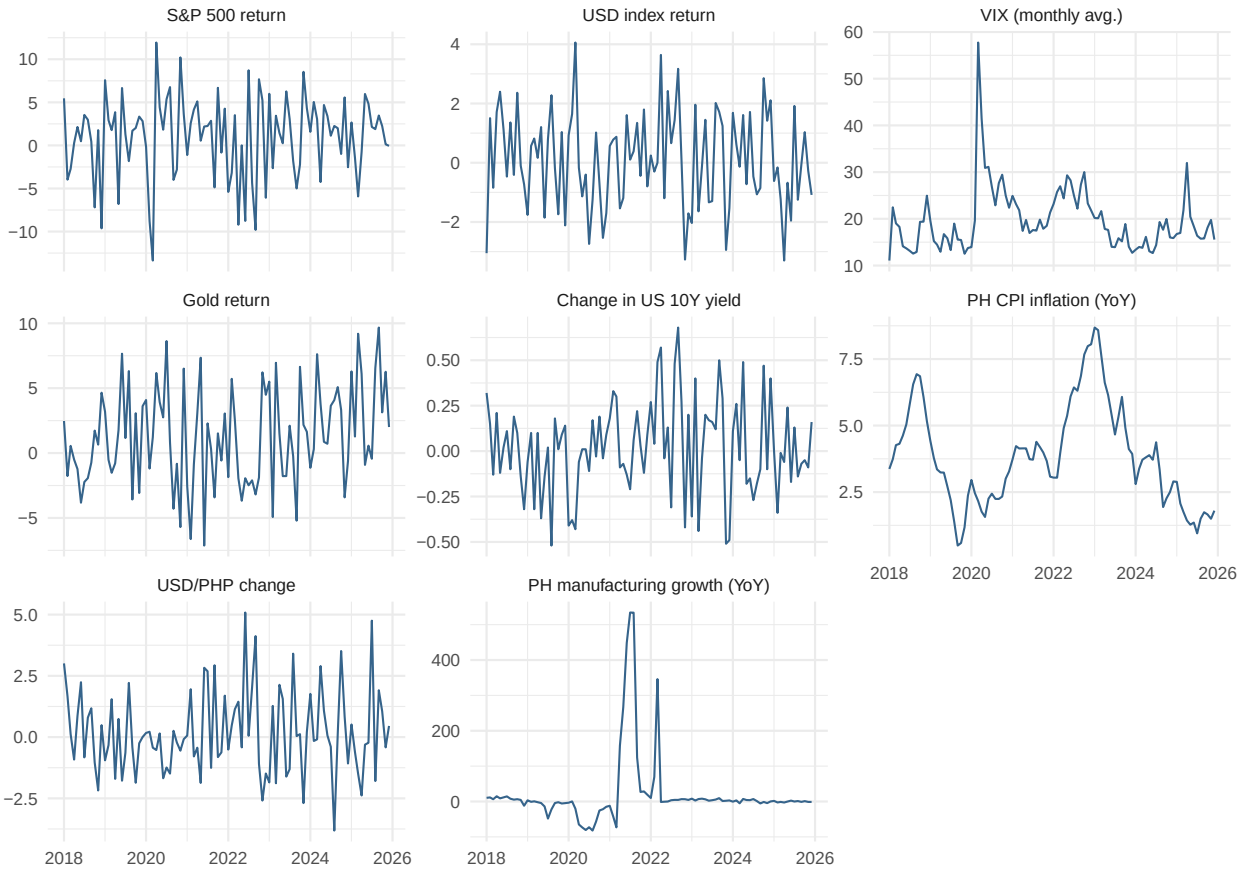



Figure 2: Explanatory variables as used in the regression

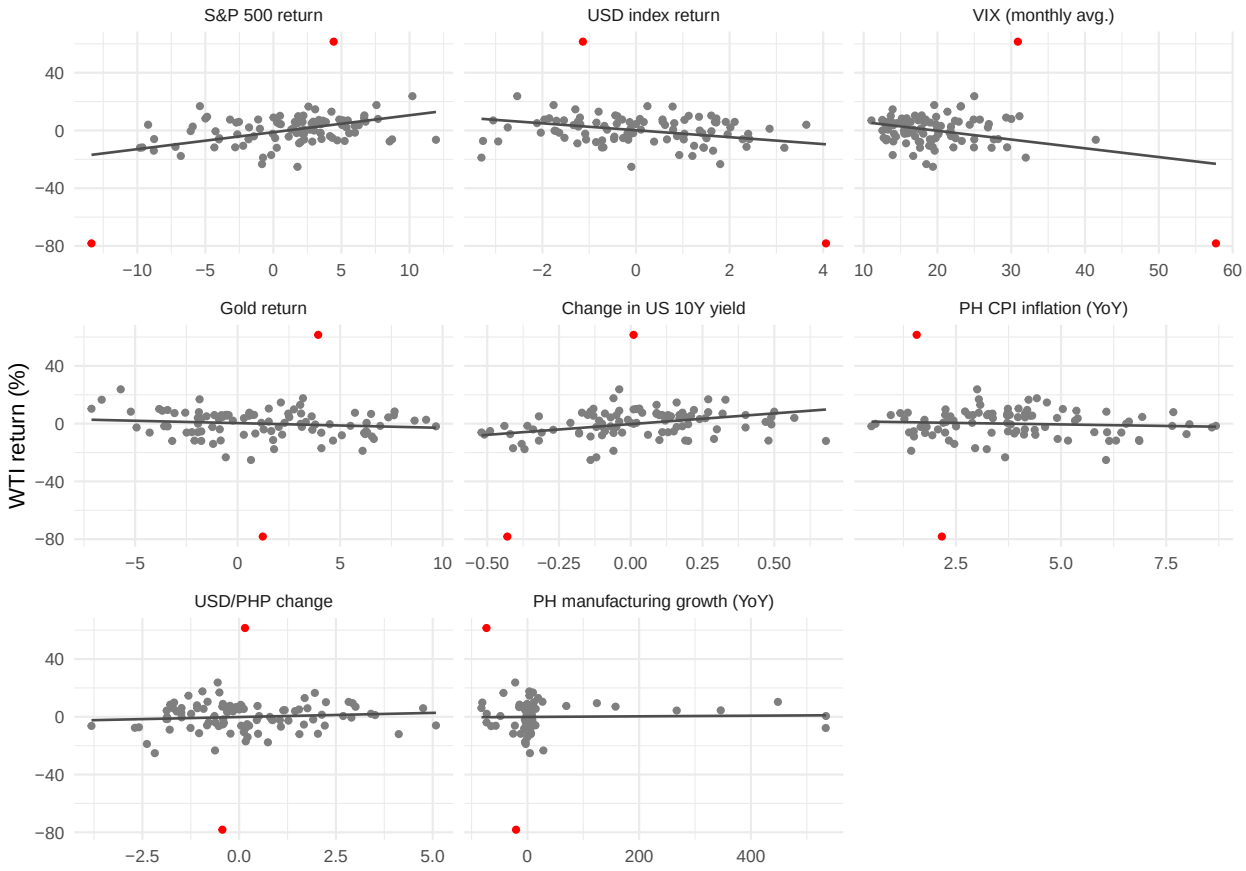



Figure 3: WTI return against each candidate regressor (OLS line; March and May 2020 in red)

Figure 1 shows the 2020 collapse and rebound, the 2022 spike after Russia's invasion of Ukraine, and a gradual decline from 2023. Figures 2 and 3 show the main patterns. WTI tends to rise with equities and US yields and to fall when the dollar strengthens or volatility rises. No clear relationship appears with any Philippine variable. The two 2020 outliers (red) sit far from every other point and could pull the fitted lines on their own.

## 2.4 Transformations and exclusions

- **Log returns** for prices and indices (WTI, S&P 500, USD index, gold, USD/PHP) make the series approximately stationary and comparable in percentage terms. **First differences** are used for the 10-year yield because it is already in percent. **Levels** are used for VIX, and **YoY rates** for PH CPI and manufacturing, as the brief specifies.
- **No observations are excluded.** The extreme months of 2020 are real market outcomes. They are kept in the sample and dealt with explicitly in Sections 6–8.
- The **change in VIX** (`dVIX`) is constructed as a possible alternative to the VIX level and is assessed in Section 5.

# 3 Initial Multiple Regression Model

## 3.1 Specification

The baseline model includes all eight candidate regressors from the brief. Each one has a clear theoretical link to oil prices: demand and growth expectations (S&P 500, US yields), the dollar pricing of oil (USD index), risk sentiment (VIX), safe-haven and inflation-hedge flows (gold), and the Philippine block:

$$\begin{aligned}\mathrm{WTI}_t = \beta_0 + \beta_1\,\mathrm{SP500}_t + \beta_2\,\mathrm{USD}_t + \beta_3\,\mathrm{VIX}_t + \beta_4\,\mathrm{Gold}_t + \beta_5\,\Delta 10\mathrm{Y}_t \\ + \beta_6\,\mathrm{PH_CPI}_t + \beta_7\,\mathrm{USDPHP}_t + \beta_8\,\mathrm{PH_Mfg}_t + u_t\end{aligned}$$

The model is estimated by ordinary least squares (OLS) on 96 monthly observations. All regressors are measured in the same month as the WTI return. The model therefore describes **contemporaneous associations**: how oil moves together with financial and economic conditions in the same month. It is not a forecasting model and does not identify causal effects.

## 3.2 Estimation results

Table 4: Initial model (M1): OLS estimates with classical standard errors

| Term | Estimate | Std. Error | t value | p-value | 95% CI low | 95% CI high |
|---|---|---|---|---|---|---|
| Intercept | 8.273 | 4.543 | 1.821 | 0.072 | -0.756 | 17.301 |
| S&P 500 return | 0.866 | 0.296 | 2.924 | 0.004 | 0.277 | 1.455 |
| USD index return | -3.120 | 1.036 | -3.012 | 0.003 | -5.180 | -1.061 |
| VIX (monthly avg.) | -0.355 | 0.174 | -2.045 | 0.044 | -0.701 | -0.010 |
| Gold return | -0.413 | 0.355 | -1.163 | 0.248 | -1.118 | 0.293 |
| Change in US 10Y yield | 18.541 | 4.799 | 3.864 | <0.001 | 9.002 | 28.079 |
| PH CPI inflation (YoY) | -0.510 | 0.605 | -0.843 | 0.401 | -1.712 | 0.692 |
| USD/PHP change | 1.690 | 0.828 | 2.042 | 0.044 | 0.045 | 3.336 |
| PH manufacturing growth (YoY) | -0.003 | 0.011 | -0.257 | 0.798 | -0.025 | 0.019 |

Table 5: Initial model (M1): goodness of fit and overall F-test

| Model | N | k | R2 | Adj. R2 | Resid. SE | F | F p-value | AIC | BIC |
|---|---|---|---|---|---|---|---|---|---|
| M1 | 96 | 8 | 0.412 | 0.358 | 10.869 | 7.609 | <0.001 | 741.076 | 766.719 |

The model explains 41.2% of the monthly variation in WTI returns ($\bar{R}^2 = 0.358$). The overall F-test of $H_0 : \beta_1 = \cdots = \beta_8 = 0$ gives $F(8, 87) = 7.61$ with $p < 0.001$. The regressors are therefore jointly associated with oil returns. A residual standard error of 10.87 percentage points shows that much of the monthly movement in oil is left unexplained.

## 3.3 Coefficient interpretation (ceteris paribus)

Each coefficient gives the expected change in the monthly WTI log return, in percentage points, for a one-unit change in that regressor, *holding all other regressors constant*:

- **S&P 500 (0.866, p = 0.004).** A 1% higher equity return goes with a 0.87 pp higher oil return. Rising equities signal stronger expected global demand.

- **USD index (-3.120, p = 0.003).** A 1% appreciation of the broad dollar goes with oil returns 3.12 pp lower. Oil is invoiced in dollars, so a stronger dollar makes it more expensive for non-US buyers and lowers demand.
- **VIX (-0.355, p = 0.044).** Each extra point of average implied volatility goes with a 0.36 pp lower oil return. This is consistent with risk-off episodes lowering demand for risky commodities.
- **Gold (-0.413, p = 0.248).** The partial association with gold is negative but not significant once the dollar and risk are controlled for.
- **US 10Y yield (18.541, p < 0.001).** A 10 basis-point rise in the 10-year yield goes with oil returns 1.85 pp higher. Rising yields reflect stronger growth and inflation expectations, which are also positive for oil demand.
- **PH CPI (-0.510, p = 0.401) and PH manufacturing (-0.0028, p = 0.798).** Neither is statistically different from zero.
- **USD/PHP (1.690, p = 0.044).** Peso depreciation goes with higher oil returns. The economic direction is almost certainly *from oil to the peso*: the Philippines imports its oil, so higher prices widen the trade deficit and weaken the peso.

### 3.4 Economic plausibility

Table 6: Expected versus estimated signs, model M1

| Variable | Expected | Estimated | Significant at 5% |
|---|---|---|---|
| S&P 500 return | + | + | Yes |
| USD index return | - | - | Yes |
| VIX (monthly avg.) | - | - | Yes |
| Gold return | +/- | - | No |
| Change in US 10Y yield | + | + | Yes |
| PH CPI inflation (YoY) | +/0 | - | No |
| USD/PHP change | +/0 | + | Yes |
| PH manufacturing growth (YoY) | +/0 | - | No |

The signs of the global variables match theory. The magnitudes are also reasonable. Oil is roughly three times as volatile as the S&P 500, so an equity beta below one is plausible. The yield coefficient looks large but works out to about 1.85 pp per 10 bp. The Philippine variables have no clear relationship with WTI, which fits the Philippines being a price-taker. The significant USD/PHP coefficient is better read as oil moving the peso than the reverse. Two cautions apply before relying on these numbers. First, the descriptive analysis showed two extreme months that may dominate OLS. Second, classical standard errors assume homoskedastic, serially uncorrelated errors. Both points are examined below.

## 4 Statistical Inference

### 4.1 Individual t-tests

For each regressor the test is $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$, using $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$ with 87 degrees of freedom. Two-sided tests are used even where theory suggests a sign, so that the analysis does not assume the answer. A one-sided test for the dollar ($H_1 : \beta_{USD} < 0$) is reported as well, because the dollar-pricing argument makes a clear directional prediction.

Using two-sided tests at the 5% level, H0 is rejected for the S&P 500, the dollar, VIX, the 10-year yield and USD/PHP. It is not rejected for gold, PH inflation or PH manufacturing. The one-sided test for the dollar gives $p = 0.0017$, which strongly supports the expected negative effect.

Table 7: t-tests of $H_0 : \beta_j = 0$, model M1

| Variable | t value | p-value (two-sided) | Decision at 5% |
|---|---|---|---|
| S&P 500 return | 2.924 | 0.004 | Reject H0 |
| USD index return | -3.012 | 0.003 | Reject H0 |
| VIX (monthly avg.) | -2.045 | 0.044 | Reject H0 |
| Gold return | -1.163 | 0.248 | Do not reject H0 |
| Change in US 10Y yield | 3.864 | <0.001 | Reject H0 |
| PH CPI inflation (YoY) | -0.843 | 0.401 | Do not reject H0 |
| USD/PHP change | 2.042 | 0.044 | Reject H0 |
| PH manufacturing growth (YoY) | -0.257 | 0.798 | Do not reject H0 |

## 4.2 Confidence intervals

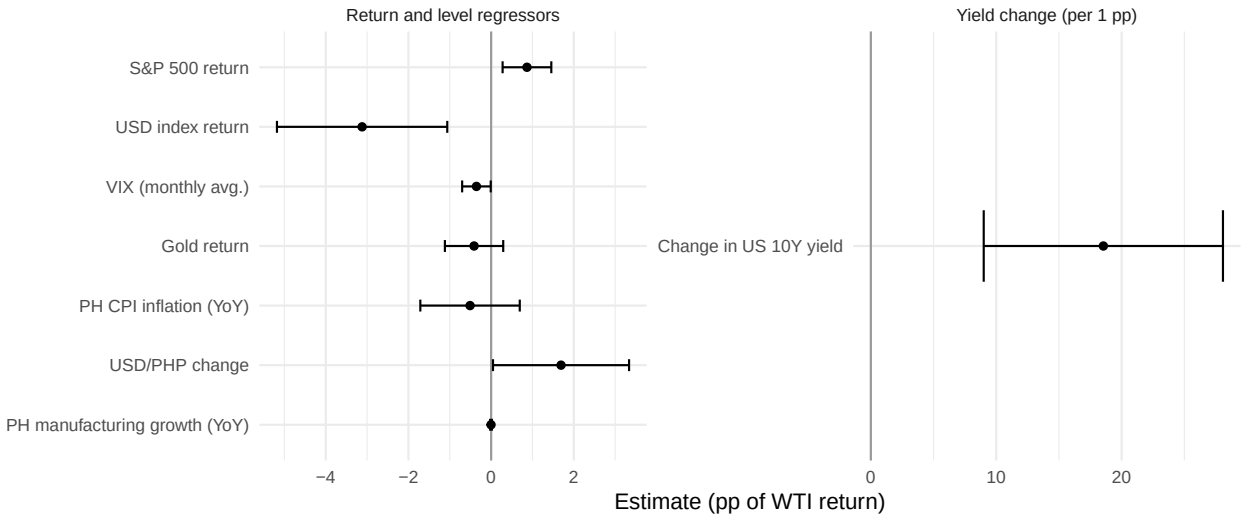



Figure 4: 95% confidence intervals for the M1 slope coefficients

The confidence intervals in the M1 coefficient table and in Figure 4 show how uncertain the estimates are. The S&P 500 interval runs from 0.28 to 1.46, so the data are consistent with oil moving anywhere from a quarter to almost one-and-a-half times as much as equities. The dollar interval (-5.18 to -1.06) is wide but lies entirely below zero. The VIX interval only just excludes zero. The intervals for PH inflation and PH manufacturing comfortably include zero, and the USD/PHP interval only just excludes it. With only 96 observations and very volatile oil returns, the data establish the *direction* of the global effects more firmly than their size.

## 4.3 Statistical versus economic significance

Statistical significance shows that an association is unlikely to be zero. It does not show that the association is large enough to matter. Coefficients are measured in different units, so the table below converts each one into the effect of a **one-standard-deviation** move in the regressor, expressed in pp of WTI return and as a share of the standard deviation of WTI returns (13.56 pp).

The largest effects come from the dollar, the 10-year yield and the S&P 500. A one-SD move in any of them changes the monthly oil return by roughly 4–5 pp, about a third of oil's own monthly SD. The VIX effect is about half as large. Effects of this size matter for any firm with oil exposure: on a $10 million monthly fuel bill, 4 pp is about $0.4 million. USD/PHP is statistically significant but

Table 8: Economic significance: effect of a one-standard-deviation move, model M1

| Variable | SD of X | Effect of +1 SD (pp) | % of SD(WTI) | p-value |
|---|---|---|---|---|
| S&P 500 return | 4.77 | 4.13 | 30.47 | 0.004 |
| USD index return | 1.60 | -4.98 | -36.75 | 0.003 |
| VIX (monthly avg.) | 6.76 | -2.40 | -17.73 | 0.044 |
| Gold return | 3.79 | -1.56 | -11.54 | 0.248 |
| Change in US 10Y yield | 0.26 | 4.76 | 35.13 | <0.001 |
| PH CPI inflation (YoY) | 1.91 | -0.98 | -7.19 | 0.401 |
| USD/PHP change | 1.68 | 2.85 | 21.00 | 0.044 |
| PH manufacturing growth (YoY) | 103.29 | -0.29 | -2.16 | 0.798 |

economically modest, at about 2.8 pp per SD, and as argued above it most likely reflects oil's effect on the peso. Philippine manufacturing growth has an enormous SD because of the 2020–21 base effects, yet its effect is essentially zero. **It is both statistically and economically insignificant.**

### 4.4 Joint hypothesis tests

Table 9: Joint F-tests on model M1

| Hypothesis | Restrictions | F | p-value |
|---|---|---|---|
| Philippine block: PH_CPI = USDPHP = PH_Mfg = 0 | 3 | 1.720 | 0.169 |
| Global block: SP500 = USD = VIX = Gold = d10Y = 0 | 5 | 11.907 | <0.001 |
| PH macro only: PH_CPI = PH_Mfg = 0 | 2 | 0.413 | 0.663 |

The most useful financial question for a Philippine decision-maker is whether **domestic conditions add information about oil once global conditions are known**. The joint test of the Philippine block gives $F = 1.72$ ($p = 0.169$). This is not significant: as a group, Philippine inflation, the peso and manufacturing add nothing reliable. Dropping the peso, which is significant on its own, leaves domestic inflation and manufacturing with $p = 0.663$. By contrast, the global block is overwhelmingly significant ($p < 0.001$). In financial terms, the variation in oil returns that a Philippine firm faces comes from global markets, not from the domestic economy.

## 5 Model Evaluation and Comparison

Three theory-motivated specifications are compared:

- **M1:** all eight regressors in the brief.
- **M2:** global block only, dropping the Philippine variables. This tests the price-taker hypothesis from Section 4.
- **M3:** M2 with the VIX *level* replaced by its monthly *change*. The dependent variable is a monthly change, so a change in risk perception is arguably the better-matched regressor.

**Nested test (M2 vs M1).** The partial F-test of the restriction that removes the Philippine block gives $F = 1.72$ with $p = 0.169$. The three Philippine variables can be dropped without a significant loss of fit. The information criteria agree: BIC falls from 766.7 to 758.6 and AIC from 741.1 to 740.6. M1 has the higher $R^2$, but only because adding regressors can never lower $R^2$. After adjusting for degrees of freedom the gain is only 0.015.

**Non-nested comparison (M3 vs M2).** M3 looks clearly better ($\bar{R}^2$ of 0.434 against 0.342, and AIC

Table 10: Fit comparison of candidate specifications

| Model | N | k | R2 | Adj. R2 | Resid. SE | F | F p-value | AIC | BIC |
|---|---|---|---|---|---|---|---|---|---|
| M1: brief (8 regressors) | 96 | 8 | 0.412 | 0.358 | 10.869 | 7.609 | <0.001 | 741.076 | 766.719 |
| M2: global block | 96 | 5 | 0.377 | 0.342 | 10.998 | 10.881 | <0.001 | 740.608 | 758.558 |
| M3: global, change in VIX | 96 | 5 | 0.464 | 0.434 | 10.198 | 15.593 | <0.001 | 726.099 | 744.050 |

lower by 14.5). Selecting on fit alone would be a mistake, though. When M2 and M3 are re-estimated without March and May 2020, the change-in-VIX coefficient drops from -1.13 to -0.01 ($p = 0.967$). The two models then fit almost identically ($\bar{R}^2$ of 0.241 for M2 and 0.231 for M3). M3's apparent advantage comes from **one month**: March 2020, when the VIX jumped 38 points as oil crashed. That is a coincidence of crises, not a stable relationship. M3 also weakens the equity coefficient, because the change in VIX and S&P 500 returns are strongly correlated (–0.61).

**Preferred specification at this stage: M2.** It improves on M1 in both AIC and BIC, follows directly from the joint test of the price-taker hypothesis, and keeps the VIX measure the brief specifies. Unlike M3, its fit does not rest on a single crisis observation. The diagnostics in the next section test whether M2 is adequate. They also show that M1 and M2 share a deeper problem: both are heavily influenced by the 2020 extremes.

# 6 Regression Diagnostics

Diagnostics are run on both the initial model (M1) and the preferred specification (M2), so that the remedies in Section 7 tackle problems both models share.

Table 11: Diagnostic tests (p-values unless stated; Durbin-Watson is the statistic)

| Model | Max VIF | Breusch-Pagan p | White p | Durbin-Watson | Breusch-Godfrey(12) p | Jarque-Bera p | RESET p |
|---|---|---|---|---|---|---|---|
| M1 | 2.201 | 0.102 | 0.071 | 1.793 | 0.322 | <0.001 | <0.001 |
| M2 | 1.860 | 0.048 | 0.106 | 1.856 | 0.466 | <0.001 | <0.001 |

*Note:*
Breusch-Pagan and White: H0 homoskedasticity. Breusch-Godfrey: H0 no autocorrelation up to lag 12. Jarque-Bera: H0 normal errors. RESET: H0 correct functional form.

## 6.1 Multicollinearity

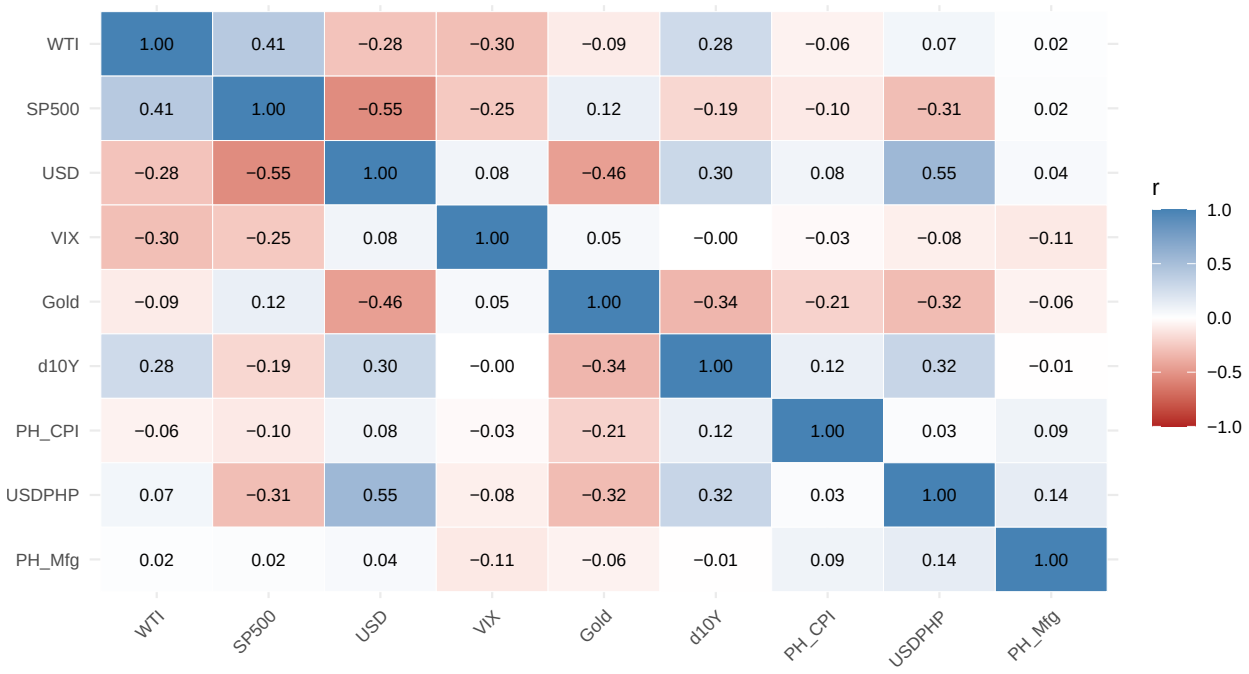



Figure 5: Correlation matrix of the dependent and explanatory variables

Table 12: Variance inflation factors, model M1

| Variable | VIF | 1/VIF |
|---|---|---|
| S&P 500 return | 1.61 | 0.62 |
| USD index return | 2.20 | 0.45 |
| VIX (monthly avg.) | 1.11 | 0.90 |
| Gold return | 1.46 | 0.69 |
| Change in US 10Y yield | 1.22 | 0.82 |
| PH CPI inflation (YoY) | 1.08 | 0.93 |
| USD/PHP change | 1.56 | 0.64 |
| PH manufacturing growth (YoY) | 1.04 | 0.96 |

The largest pairwise correlations are between the dollar and USD/PHP (0.55) and between the dollar and the S&P 500 (-0.55). All VIFs are below 3, far under the usual warning levels of 5–10. **Multicollinearity is not a concern.** The wide confidence intervals in Section 4 come from noisy oil returns, not from overlapping regressors.

## 6.2 Heteroskedasticity

For M1, the Breusch-Pagan test ($p = 0.102$) and the White test ($p = 0.071$) do not reject homoskedasticity at 5%, though White comes close at 10%. For M2, the Breusch-Pagan test rejects ($p = 0.048$). The residual-versus-fitted plot below shows why: residual spread is far larger at the extreme fitted values of early 2020. **The evidence for heteroskedasticity is moderate and mostly tied to the crisis months.**

### 6.3 Autocorrelation

These are time-series data, so serial correlation has to be checked. The Durbin-Watson statistics (1.79 and 1.86) are close to 2. The Breusch-Godfrey test up to 12 lags does not reject ($p = 0.322$ and 0.466). This is what efficient-market reasoning predicts for monthly returns, and **there is no meaningful autocorrelation.**

### 6.4 Outliers and influential observations

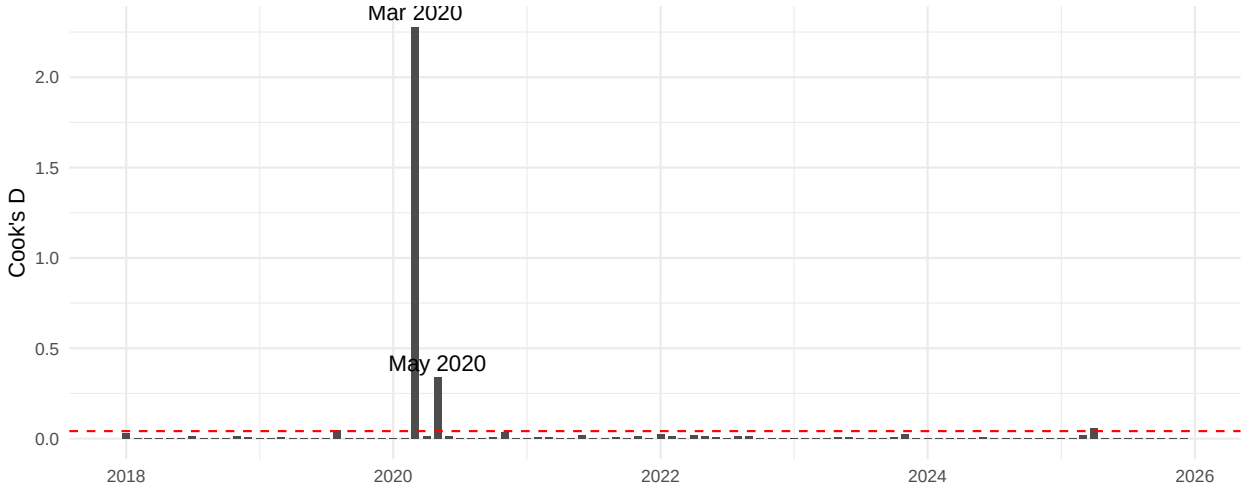



Figure 6: Cook's distance for model M2 (dashed line: 4/n)

Table 13: Most influential months, model M2

| Date | WTI | Std. residual | Leverage | Cook's D |
|---|---|---|---|---|
| Mar 2020 | -78.196 | -4.439 | 0.456 | 2.279 |
| May 2020 | 61.503 | 7.208 | 0.058 | 0.340 |
| Apr 2025 | -18.804 | -1.615 | 0.121 | 0.059 |
| Aug 2019 | -6.093 | 1.349 | 0.134 | 0.046 |
| Nov 2020 | 23.763 | 1.052 | 0.156 | 0.034 |
| Jan 2018 | 6.963 | -1.242 | 0.101 | 0.029 |

Two months dominate. **March 2020** (Cook's D = 2.28) saw the collapse of the OPEC+ agreement on 6 March and the start of COVID-19 lockdowns at the same time, and WTI fell 78.2%. **May 2020** brought a 61.5% rebound after the April futures collapse, as record OPEC+ cuts took effect. A single observation with Cook's D above 1 can by itself move the estimated coefficients substantially. Both months are also extreme outliers: the Bonferroni-adjusted outlier test gives $p < 0.001$. Neither is a data error. They are real, exceptional *supply-and-demand events* that the financial regressors cannot capture.

## 6.5 Residual diagnostics

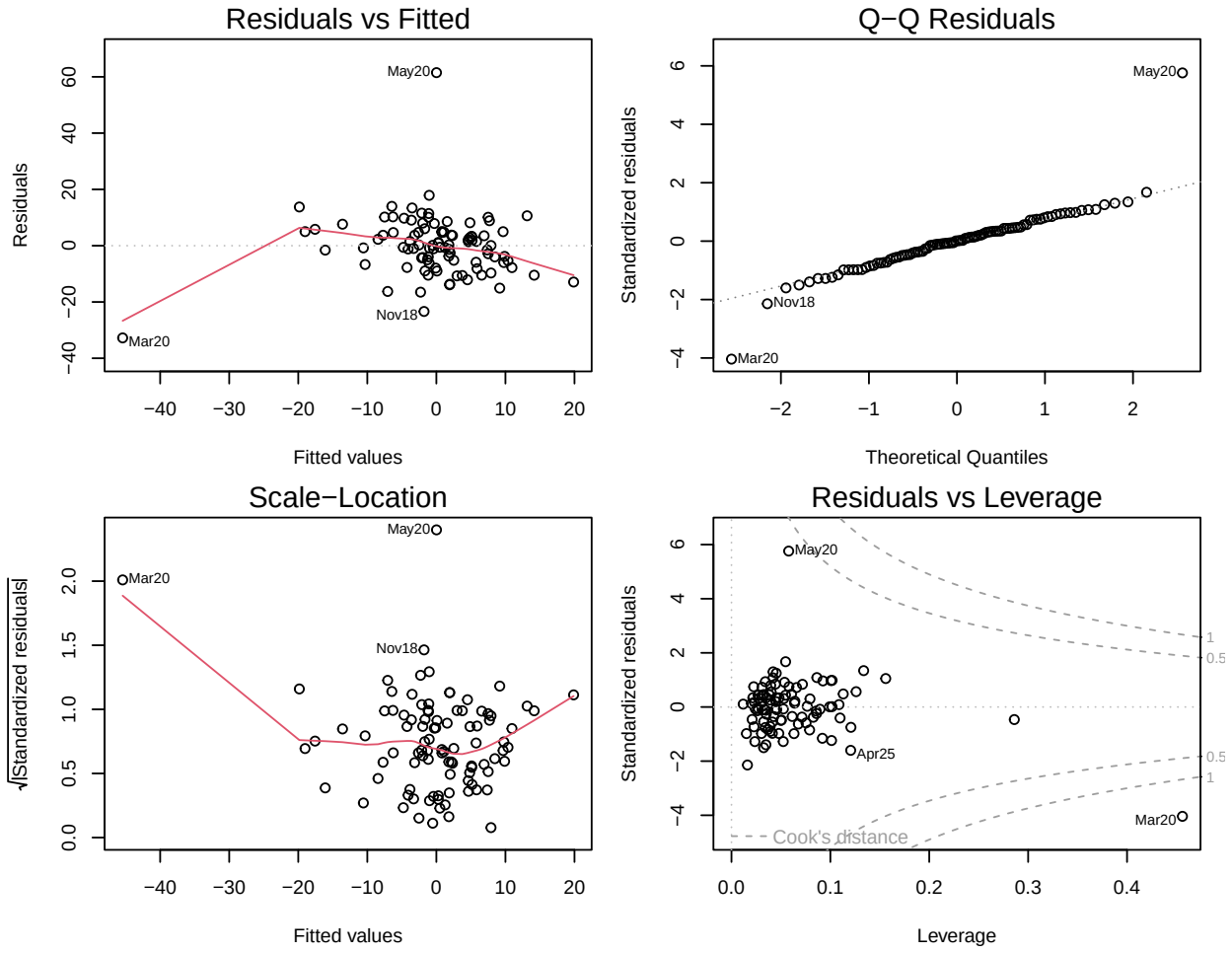



Figure 7: Residual diagnostics for model M2

The Q-Q plot shows residuals that are normal in the middle but have two extreme tails, and the Jarque-Bera test rejects normality decisively. The RESET test also rejects correct functional form ($p < 0.001$ for M2). This most likely reflects the two crisis months bending the fit rather than a true non-linearity, and Section 7 tests that directly.

## 6.6 Summary of diagnostic findings

| Issue | Evidence | Severity |
|---|---|---|
| Multicollinearity | VIF $< 2.5$, correlations $< 0.6$ | None |
| Heteroskedasticity | BP / White borderline, driven by 2020 | Moderate |
| Autocorrelation | DW about 1.8–1.9; BG(12) not significant | None |
| Outliers / influence | Mar & May 2020: Cook's D $> 1$, Bonferroni outliers | **Severe** |
| Normality / specification | Jarque-Bera and RESET reject | **Severe** (linked to the outliers) |

# 7 Model Improvement

## 7.1 Problems identified

The diagnostics point to one dominant problem and one secondary concern:

1. **Two extreme, influential months (March and May 2020).** These months alone cause the non-normal residuals and the failed RESET test, and they can distort every slope coefficient. This is the most important problem.
2. **Moderate heteroskedasticity**, concentrated in the same period. With a time-series sample, some residual serial correlation is also possible even though the tests do not detect it.

Multicollinearity and autocorrelation need no remedy. The Philippine block has already been removed through the nested F-test in Section 5.

## 7.2 Remedies applied and why

**(a) Event (pulse) dummies for March 2020 and May 2020.** Each dummy equals 1 in its month and 0 otherwise. This is a standard *intervention* specification for known, dated, exogenous shocks. The OPEC+ price war, the lockdown demand collapse and the post-April rebound were physical oil-market events that equity, dollar, volatility, gold and yield movements cannot represent. Dummies are better than deleting the months, because they keep the sample intact and make the size of each shock explicit as its own coefficient. They are also better than one combined "COVID" dummy: the crash and the rebound have opposite signs, so a single dummy would average them to nothing. Separate dummies are not added for every large residual, because only these two months are statistical outliers (Bonferroni test) with a documented, oil-specific cause.

**(b) Newey-West (HAC) standard errors** with three lags, the usual $T^{1/4}$ rule for $T = 96$. These protect the t-tests and confidence intervals against any remaining heteroskedasticity and low-order autocorrelation of unknown form. They leave the coefficient estimates unchanged. Weighted least squares is not used, because it would require a model of how the error variance behaves, and the diagnostics only suggest that the variance is high during crises. A log or other transformation of the dependent variable is also unnecessary, since WTI is already a log return.

The improved model is

$$\mathrm{WTI}_t = \beta_0 + \beta_1 \mathrm{SP500}_t + \beta_2 \mathrm{USD}_t + \beta_3 \mathrm{VIX}_t + \beta_4 \mathrm{Gold}_t + \beta_5 \Delta 10\mathrm{Y}_t + \delta_1 D_t^{Mar20} + \delta_2 D_t^{May20} + u_t$$

Table 15: Improved model (M4): OLS estimates with Newey-West HAC standard errors

| Term | Estimate | Std. Error | t value | p-value | 95% CI low | 95% CI high |
|---|---|---|---|---|---|---|
| Intercept | 2.914 | 2.240 | 1.301 | 0.197 | -1.537 | 7.364 |
| S&P 500 return | 0.609 | 0.182 | 3.344 | 0.001 | 0.247 | 0.972 |
| USD index return | -1.348 | 0.667 | -2.021 | 0.046 | -2.673 | -0.023 |
| VIX (monthly avg.) | -0.163 | 0.128 | -1.270 | 0.208 | -0.418 | 0.092 |
| Gold return | -0.469 | 0.210 | -2.231 | 0.028 | -0.886 | -0.051 |
| Change in US 10Y yield | 13.547 | 3.020 | 4.486 | <0.001 | 7.545 | 19.549 |
| Dummy: March 2020 | -51.696 | 6.263 | -8.254 | <0.001 | -64.142 | -39.250 |
| Dummy: May 2020 | 61.095 | 1.904 | 32.081 | <0.001 | 57.311 | 64.880 |

Table 16: Coefficient comparison: estimate (standard error); * p<0.10, ** p<0.05, *** p<0.01

| Term | M1 initial (OLS SE) | M2 global (OLS SE) | M4 improved (HAC SE) |
|---|---|---|---|
| Intercept | 8.273* (4.543) | 7.245* (3.707) | 2.914 (2.240) |
| S&P 500 return | 0.866*** (0.296) | 0.862*** (0.297) | 0.609*** (0.182) |
| USD index return | -3.120*** (1.036) | -2.232** (0.964) | -1.348** (0.667) |
| VIX (monthly avg.) | -0.355** (0.174) | -0.398** (0.173) | -0.163 (0.128) |
| Gold return | -0.413 (0.355) | -0.380 (0.352) | -0.469** (0.210) |
| Change in US 10Y yield | 18.541*** (4.799) | 20.146*** (4.764) | 13.547*** (3.020) |
| PH CPI inflation (YoY) | -0.510 (0.605) | | |
| USD/PHP change | 1.690** (0.828) | | |
| PH manufacturing growth (YoY) | -0.003 (0.011) | | |
| Dummy: March 2020 | | | -51.696*** (6.263) |
| Dummy: May 2020 | | | 61.095*** (1.904) |
| Adj. R2 | 0.358 | 0.342 | 0.664 |
| BIC | 766.7 | 758.6 | 701.0 |

Table 17: Diagnostics before and after improvement

| Model | Max VIF | Breusch-Pagan p | White p | Durbin-Watson | Breusch-Godfrey(12) p | Jarque-Bera p | RESET p |
|---|---|---|---|---|---|---|---|
| M1 (initial) | 2.201 | 0.102 | 0.071 | 1.793 | 0.322 | <0.001 | <0.001 |
| M2 (global) | 1.860 | 0.048 | 0.106 | 1.856 | 0.466 | <0.001 | <0.001 |
| M4 (improved) | 1.970 | 0.374 | 0.614 | 1.907 | 0.882 | 0.172 | 0.891 |

### 7.3 Comparison with the initial model

The changes are large and lead to useful conclusions:

- **The specification problems are resolved.** In M4 the RESET test no longer rejects ($p = 0.891$), residuals are consistent with normality (Jarque-Bera $p = 0.172$), and neither heteroskedasticity test rejects (BP $p = 0.374$, White $p = 0.614$). The earlier failures came from two months, not from a misspecified functional form.
- **Magnitudes fall towards more realistic values.** The S&P 500 coefficient falls from 0.87 to 0.61, the dollar from -3.12 to -1.35 and the yield from 18.5 to 13.5. The 2020 extremes had inflated every global coefficient, because markets fell together in March 2020 and recovered together afterwards.
- **Significance changes.** The S&P 500 and the 10-year yield remain significant under HAC inference. The dollar ($p = 0.046$) and gold ($p = 0.028$) are now significant at 5%. **VIX loses significance** ($p = 0.208$): its apparent effect in M1 depended on the crisis months.
- **The Philippine variables still add nothing.** When they are added back to M4, the HAC Wald test gives $F = 1.96$, $p = 0.126$.
- **Interpret the jump in $\bar{R}^2$ (to 0.664) with care.** Much of it is mechanical, because the two dummies fit their own months exactly. M4 should be judged on the improved diagnostics and the more stable coefficients, not on $R^2$.

**M4 is the selected final model.** One concern remains: the dummies were chosen after looking at the data. Section 8 therefore checks the result with a method that does not require choosing which months to treat specially.

# 8 Advanced Regression Component: Robust MM-Estimation

## 8.1 Why robust regression?

Robust regression is chosen because the diagnostics call for it. Monthly oil returns are fat-tailed, and Section 6 showed that a couple of extreme months can control OLS estimates. The dummy remedy in Section 7 works, but it depends on the analyst picking which months to neutralise *after* seeing the residuals, which risks data snooping. A firm using this model for risk management also needs to know whether the global sensitivities hold up without hand-picked adjustments.

The **MM-estimator** (Yohai, 1987) addresses this directly:

- It replaces OLS's squared-error loss with a bounded loss (Tukey's bisquare). Each observation's influence is therefore capped, and observations with very large residuals receive zero weight.
- It starts from a high-breakdown S-estimate, so up to 50% of the data could be contaminated without the estimates breaking down. It is then tuned to be 95% as efficient as OLS when errors really are normal.
- It **chooses the weights from the data**, so no dates need to be specified.

The method is applied to the M2 regressors with **no event dummies**, to see whether it reaches the same answer as M4 on its own. Other advanced methods were considered and not used:

- Logistic regression and panel methods do not fit a single continuous time series.
- Regularisation is not needed with five regressors and no multicollinearity.
- An interaction or regime model (for example, high- versus low-VIX months) would split an already small sample of 96 months.

## 8.2 Results

Table 18: Robust MM-estimates compared with OLS: estimate (standard error); * p<0.10, ** p<0.05, *** p<0.01

| Term | M2 OLS (no dummies) | M4 OLS + dummies (HAC) | MM robust (no dummies) |
|---|---|---|---|
| Intercept | 7.245* (3.707) | 2.914 (2.240) | 2.989 (2.718) |
| S&P 500 return | 0.862*** (0.297) | 0.609*** (0.182) | 0.598*** (0.218) |
| USD index return | -2.232** (0.964) | -1.348** (0.667) | -1.265* (0.707) |
| VIX (monthly avg.) | -0.398** (0.173) | -0.163 (0.128) | -0.151 (0.127) |
| Gold return | -0.380 (0.352) | -0.469** (0.210) | -0.472* (0.258) |
| Change in US 10Y yield | 20.146*** (4.764) | 13.547*** (3.020) | 12.872*** (3.492) |
| Dummy: March 2020 | | -51.696*** (6.263) | |
| Dummy: May 2020 | | 61.095*** (1.904) | |

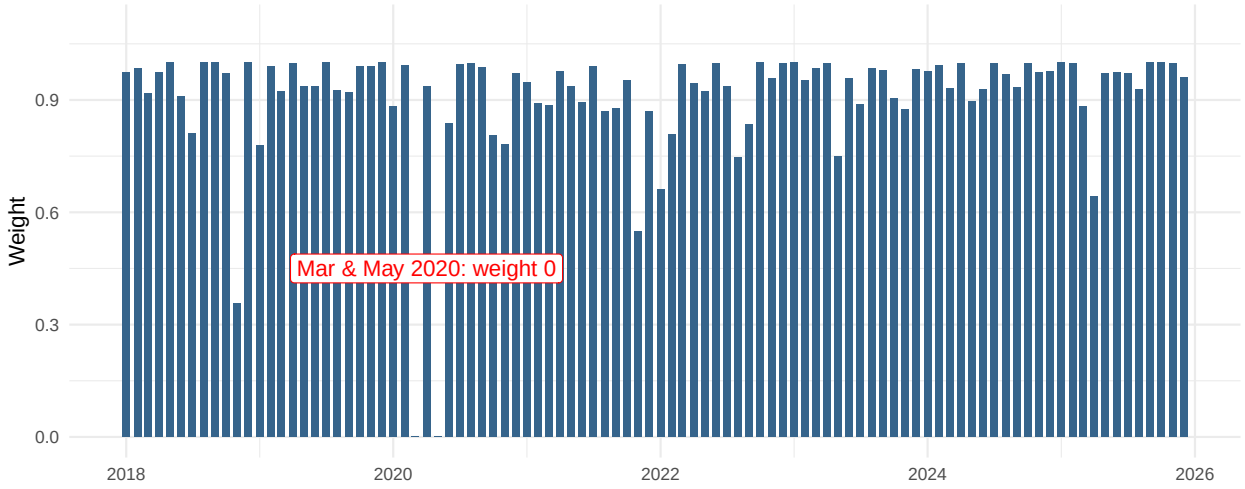



Figure 8: Final MM-estimation weight of each month (1 = full weight, 0 = ignored)

Table 19: Months down-weighted below 0.5 by the MM-estimator

| Date | WTI | Weight |
|---|---|---|
| Mar 2020 | -78.196 | 0.000 |
| May 2020 | 61.503 | 0.000 |
| Nov 2018 | -25.164 | 0.356 |

## 8.3 Interpretation

The robust estimator independently picks out **exactly the same two months**. March 2020 and May 2020 receive weights of 0.000 and 0.000, so they are effectively excluded. Apart from these two, only 1 month(s) receive a weight below 0.5 (November 2018), and none is excluded. The robust coefficients are close to those of the dummy model M4:

- S&P 500: 0.60 (robust) vs 0.61 (M4)
- USD index: -1.26 vs -1.35
- Gold: -0.47 vs -0.47
- US 10Y yield: 12.9 vs 13.5
- VIX: -0.15 vs -0.16, insignificant in both

This has two consequences for the analysis:

1. **The final model is not an artefact of hand-picked dummies.** A method that knows nothing about COVID-19 or OPEC+ reaches the same conclusion, which greatly strengthens confidence in M4.
2. **The ranking of drivers is robust.** In both approaches the S&P 500 and the 10-year yield are the strongest and most reliable associations, the dollar is negative and borderline-to-significant, and VIX adds nothing once extreme months are neutralised.

M4 is kept as the reported final model because its coefficients and HAC tests are easier to communicate. The MM-estimator serves as its robustness check. One caveat applies: MM standard errors are asymptotic, so with 96 observations the robust p-values should be read as approximate.

# 9 Financial Interpretation

## 9.1 Main empirical findings

Table 20: Final model (M4): coefficients and economic significance of global drivers

| Variable | Coefficient | Effect of +1 SD (pp) | % of SD(WTI) | p-value (HAC) |
|---|---|---|---|---|
| S&P 500 return | 0.61 | 2.91 | 21.43 | 0.001 |
| USD index return | -1.35 | -2.15 | -15.87 | 0.046 |
| VIX (monthly avg.) | -0.16 | -1.10 | -8.12 | 0.208 |
| Gold return | -0.47 | -1.78 | -13.10 | 0.028 |
| Change in US 10Y yield | 13.55 | 3.48 | 25.67 | <0.001 |

1. **Oil returns move with global growth expectations.** In the final model, a one-SD rise in US 10-year yields (26 bp) goes with a 3.5 pp higher monthly WTI return, and a one-SD equity rally goes with 2.9 pp. These are the two largest and most reliable associations, and they survive both the dummy and the robust specifications.
2. **A stronger dollar goes with weaker oil.** Each 1% dollar appreciation is associated with about 1.3 pp lower WTI returns, which is consistent with oil being priced in dollars.
3. **Once the 2020 extremes are controlled for, risk sentiment (VIX) adds no explanatory power and gold has a modest negative partial association.** The negative gold coefficient fits gold and oil competing for flows during stress: when investors move into gold for safety, risky commodities sell off.
4. **Philippine conditions do not explain WTI.** Inflation, manufacturing and the peso are jointly insignificant in both the initial and the final model. The significant peso coefficient in M1 reflects causation in the other direction: oil price increases weaken the peso.
5. **Tail events dominate oil risk.** March and May 2020 moved WTI by -52 and 61 pp beyond what financial conditions predicted. That is several times the effect of any regular driver.

## 9.2 Financial and economic meaning

For a Philippine oil-consuming firm, such as a fuel distributor, an airline, a power generator or a transport company, **the risk in its oil costs is imported**. Monthly oil returns line up with US equities, US yields and the dollar, and not with anything happening in the Philippine economy. Two practical implications follow:

- **Oil and the peso compound each other.** The Philippines imports its oil, so oil price increases tend to come with peso depreciation. The peso cost of fuel rises by *more* than the dollar price of oil. A peso-based fuel budget faces both risks at once.
- **Oil shocks are a global risk factor.** The positive link with equities and yields means oil prices tend to rise when the world economy is strong. A firm whose own revenues are domestic gets no natural hedge from that global growth.

The model explains about 38% of monthly oil variation without the event dummies. The remaining variation comes from oil-specific supply news (OPEC+ decisions, inventories, geopolitics) that financial variables do not capture. Global indicators are useful for understanding the *direction* of oil risk, but **they cannot be relied on to forecast or fully hedge it**.

### 9.3 Limitations and remaining uncertainty

- **Association, not causation.** All variables are measured in the same month, and oil, equities and yields affect one another. The coefficients describe co-movement and should not be read as causal effects.
- **No forecasting ability is claimed.** Contemporaneous regressors are not known in advance. A forecasting model would need lagged regressors and an out-of-sample evaluation.
- **Small sample and structural change.** 96 months cover several distinct regimes: pre-COVID, the pandemic, the 2022 energy shock and monetary tightening. The coefficients are averages across these regimes and may not hold in the next one.
- **Event dummies are chosen after the fact.** The robust MM-estimator confirms them, but the next crisis will not come with a dummy, and tail risk remains large.
- **Data substitutions.** Gold and USD/PHP use market proxies for the WGC and BSP series, and PSA's manufacturing growth series has severe base effects. None of these affects the main conclusions, since the Philippine block is insignificant on any construction and gold is a minor driver.

## 10 Recommendation and Conclusion

**Recommendation.** A Philippine firm or treasury with significant oil exposure should:

1. **Base oil-risk monitoring and hedging on global indicators, not domestic ones.** Track US 10-year yields, US equity performance and the broad dollar index as early warning of oil cost pressure. Philippine CPI and manufacturing data say nothing about where oil is going and should not drive hedging decisions.
2. **Hedge oil and the peso together.** Because oil increases and peso weakness tend to coincide, layered WTI or Brent swaps or options should be paired with USD/PHP forward cover so that the *peso* cost of fuel is protected.
3. **Use option-based protection and stress tests for tail risk.** The largest monthly moves (–78% and +62% in 2020) were not explained by financial conditions and were several times larger than any normal-month driver. Options or collars that cap losses during extreme moves are more suitable than relying on regression-based hedge ratios alone. Budgets should be stress-tested against moves of ±50% in a single month.

**Conclusion.** Over January 2018 – December 2025, monthly WTI returns are significantly associated with **global** financial conditions: positively with equity returns and US yield changes, and negatively with the dollar. They have no reliable relationship with Philippine inflation, the peso or manufacturing growth. The initial model's results were distorted by two extreme 2020 months. Correcting for them with event dummies and HAC inference, and confirming the result with robust MM-estimation, gives a smaller but more reliable set of global sensitivities. The evidence supports a hedging policy for Philippine oil users that is driven by global markets, covers both oil and the peso, and protects against tail events.

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**Data sources:** Federal Reserve Bank of St. Louis (FRED); Philippine Statistics Authority (OpenSTAT); Yahoo Finance.